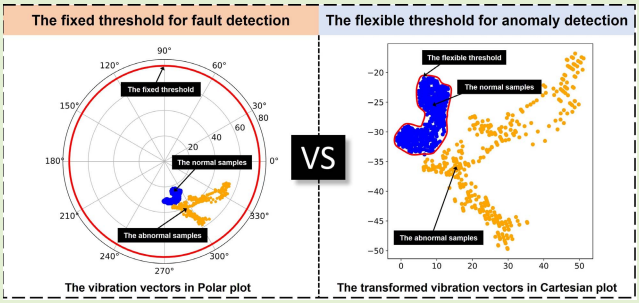
An Intelligent Anomaly Detection Method for Rotating Machinery Based on Vibration Vectors

Di Hu, Chen Zhang, Tao Yang, and Gang Chen

***Abstract*—Anomaly detection is a common technology for condition monitoring that is important for safe and reliable production in industry. In this work, we proposed an anomaly detection method for rotating machinery based on vibration vectors, which was inspired by Polar plot in vibration analysis. The vibration vector, consisting of amplitude and phase, was selected as monitored indicator at different characteristic frequencies (CFs). The original vibration vectors obtained by fast Fourier transform-based order analysis (FFT-OA) were mapped from Polar to Cartesian plot for simplified calculation and better visualization. The acceptance region defined to distinguish anomalies from normal conditions was learned by support vector data description (SVDD) model with normal vibration samples. We proposed the anomaly score to indicate the degree of deviation from normal condition, and the fit degree to reduce underfitting of data description model. Two datasets, obtained from a real steam turbine and a real steam feed pump respectively, were used to validate the method under constant and variable speed operation conditions. Compared to the amplitude-based methods, the proposed method exhibited an extremely strong ability to detect anomalies in advance. Moreover, SVDD performed better than the minimum covariance determinant (MCD) and isolated forest (iForest) methods in terms of describing the acceptance region in the present work. With low time-consuming calculation, the acceptance region can be self-evolving easily and has a prospect to the edge computing for machine condition monitoring.**



***Index Terms*—Condition monitoring, anomaly detection, rotating machinery, vibration signal, vibration vector, support vector data description.**

Abbreviations

FFT-OA	fast Fourier transform-based order analysis
DFT	discrete Fourier transform
STFT	short-time Fourier transform
CF	characteristic frequency
ANN	artificial neural network
SVDD	support vector data description
MCD	minimum covariance determinant
iForest	isolated forest
RMS	root mean square
CSO	constant speed operation
VSO	variable speed operation

Manuscript received 29 March 2022; revised 30 May 2022; accepted 30 May 2022. Date of publication 17 June 2022; date of current version 14 July 2022. This work was supported by the National Natural Science Foundation of China (NSFC) under Grant U20A20303. The associate editor coordinating the review of this article and approving it for publication was Prof. Ruqiang Yan. *(Corresponding author: Tao Yang.)*

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Digital Object Identifier 10.1109/JSEN.2022.3179740

I. Introduction

ROTATING machinery is widely used in power plant, such as steam turbine, induced draft fan, and circulating pump, the safe and reliable operation of which is vital for production benefit [1], [2]. Hence, condition monitoring has always been an important part of the lifecycle management of rotating machinery and remains a focus of research. In recent years, anomaly detection as a hot technology in condition monitoring has emerged as a new trend, which has been served to transform traditional corrective maintenance into condition-based maintenance strategy [3]–[8]. Hawkins [9] defined an anomaly as an observation generated by a different mechanism that deviates significantly from other observations, and anomalies can be detected by an acceptance region learned from normal observations. Vibration monitoring is the most reliable method for assessing the overall health and condition changes of a rotor system [10]. Thus, vibration signal-based anomaly detection is studied with two key issues. One of the issues is to extract concise and effective features from raw vibration signals, and the other is to define the acceptance

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region which can distinguish anomalies from normal condition.

In general, the condition monitoring process is divided into three steps: (1) feature extraction, (2) feature selection, and (3) classifier establishment. Several feature extraction methods have been proposed to explore the useful information from vibration signals, which can be divided into time domain analysis [11]–[20], frequency domain analysis [21]–[26], and time–frequency domain analysis [27]–[35]. Artificial neural network (ANN) methods have been introduced in recent years [36]–[40]. The feature selection is related to classifier and follows the principle of conciseness and effectiveness. Ge *et al.* [13] proposed a dynamic statistical approach to detect faults in non-Gaussian vibration systems. Purarjomandlangrudi *et al.* [18] used kurtosis and the non-Gaussianity score as features, and a Gaussian distribution-based classifier was adopted. Igba *et al.* [19] used the root mean square (RMS) and peak values of vibration signals as monitored indicators for early detection based on the extreme value theory. Su *et al.* [41] adopted the short-time Fourier transform (STFT) to obtain amplitudes at different frequencies; the classifier was established with an ANN. Tax *et al.* [42] used the RMS, kurtosis, and crest-factor as features, and the support vector data description (SVDD) was used to learn the boundary between normal and abnormal signals. Jiang *et al.* [43] used RMS as a vibration feature and established an SVDD-based classifier for a dual-rotor machine. Sadooghi *et al.* [44] performed dimensionality reduction with statistical indicators which were obtained from time and time–frequency domain, an SVDD-based classifier was used. In the present study, many schemes of fault diagnosis classifiers have been proposed to diagnose the fault directly. The fault diagnosis classifier is a multi-classifier owing to the various fault types, which the learning process requires both normal and various fault samples. However, rotating machinery operates under normal conditions in the most of running time and has various fault types and rare fault instances, which makes multi-classifier difficult to train in some situations. The anomaly detection classifier is a single-classifier and sends abnormal alarms in the early stage of fault (without knowing the type), and the learning process only requires normal samples. The arisen abnormal alarms remind the operator to take intervention measures to ensure safe and reliable running of machines. Thus, anomaly detection classifier can provide an acceptance region to detect anomalies in time and has potential in the field of condition monitoring for major machines.

The widely used anomaly detection classifier in actual production is mainly based on fixed threshold [45] of monitored indicators such as amplitude. Fixed threshold, set artificially based on experience, has an obvious shortcoming which causes either frequent alarms with a low threshold or missing of best maintenance opportunity with a high threshold. The vibration vector, consisting of amplitude and phase, contains comprehensive information about vibration component at characteristic frequencies (CFs) and is a concise and effective monitored indicator. The Polar plot is a typical vibration data presentation method [10], on which a vibration vector at any point in time is plotted as a single point. Bently [46] proposed a method based on Polar plot to detect early rotor cracks. A few studies [47]–[50] introduce vibration vector into condition monitoring for rotating machinery, in which the acceptance region on the Polar plot is set by experience or not defined. The anomaly is simply judged by the boundary with ignoring the degree of deviation from the region, which makes the approach lack robustness in industrial applications. Meanwhile, in the lifecycle of machine, the location of vibration vector may change in normal conditions, which requires the method has self-evolving ability. Consequently, the flexible and automatic description of acceptance region deserves a further study.

In this study, we adopt fast Fourier transform-based order analysis (FFT-OA) to obtain vibration vectors from raw vibration signal and use SVDD model to learn the acceptance region at different CFs. The phase information is often ignored in on-line monitoring, which may miss some signs about early fault. The amplitude and phase are jointly analyzed in a vibration vector which is mapped from Polar to Cartesian plot for simplified calculation and better visualization. We proposed the anomaly score to indicate the degree of deviation from the acceptance region, when SVDD model with Gaussian kernel function was used. Meanwhile, we proposed fit degree to solve the underfitting problem in the field of single-classifier. The other two methods minimum covariance determinant (MCD) and isolation forest (iForest) were taken for comparation. The possible anomaly can be detected when the vibration vectors are out of acceptance region with high anomaly score. Two datasets, obtained from a real steam turbine and a real steam feed pump respectively, are used to validate the method under constant speed operation (CSO) and variable speed operation (VSO) conditions. The dataset contains vibration signals of bearing, which is obtained by synchronous integrated period sampling. The case studies show that the proposed method performs better than amplitude-based method in early stage anomaly detection. The study tries to contribute in follows:

1. The practice dataset shows that acceptance region can distinguish the abnormalities from normal conditions, and the acceptance region is learned through intelligent data description methods rather than human experience.
2. The anomaly score is proposed to express the degree of deviation from the acceptance region, which provides a robustness monitoring in industrial applications.
3. The fit degree is defined to indicate the underfitting problem in single-classifier, which ensures the low rate of missing alarm in monitoring task.

The rest of this paper is organized as follows: Section II presents the intelligent anomaly detection method for rotating machinery in detail. In Section III, the actual case studies validate the proposed method. Conclusions and future work are presented in Section IV.

II. Methods

A. Acquisition of Vibration Vectors

The FFT-OA [29] technique has been widely applied in machine health monitoring, which requires synchronous integrated period sampling of the vibration signals. In the

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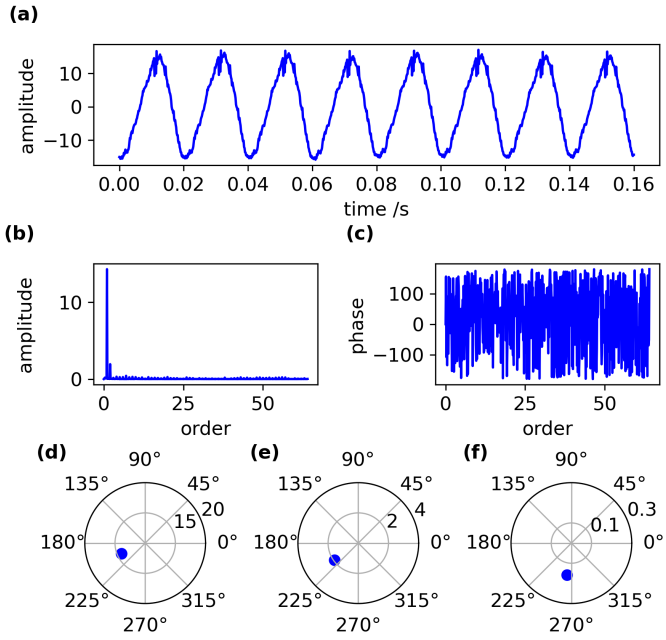


Fig. 1. FFT-OA case of a vibration signal: (a) original vibration signal; (b) amplitude spectrum; (c) phase spectrum; (d), (e), (f) the vibration vector of 1X, 2X, 3X are plotted in Polar plot respectively.

engineering, the vibration signals are regarded as discrete periodic signals. The discrete Fourier transform (DFT) can convert a vibration signal from its original time domain to a representation in the frequency domain, which decompose a time-sequence signal into CFs (fundamental and harmonics). The FFT rapidly computes such transformations by factorizing the DFT matrix into a product of sparse factors with the complexity of computing the DFT from $O\left(N^2\right)$ to $O\left(Nlog_2^N\right)$, where N is the length of the signal.

An FFT-OA case is shown in Fig. 1, in which Fig. 1a is the original vibration signal, Fig. 1b and Fig. 1c are amplitude spectrum and phase spectrum respectively. The abscissa of the amplitude and phase spectrums is order (described as $1X, 2X, \ldots$) when the vibration signal is obtained by synchronous integrated period sampling. In the Polar plot, the vibration amplitude increases from the centre, and the phase angle increases from a zero-line. A vibration vectors contains both amplitude and phase information and is represented as a point in polar plot, as shown in Fig. 1d, Fig. 1e, and Fig. 1f for $1X, 2X, 3X$, respectively. For example, when the rotor is unbalanced, the $1X$ is dominant, and the amplitude increases with the increase of speed; when the rotor is misaligned, the $2X$ component appears. Thus, the monitoring of different CFs can provide fault information.

Previous studies have focused less on the phase changes at different CFs in practice on-line monitoring. In early stage of some faults, we find that the amplitude does not change obviously, while the phase changes significantly (see Section III for details). The location of vibration vectors shows a certain distribution in Polar plot when the machine is in normal condition. Therefore, the acceptance region of different CFs can be described to detect anomalies.

B. Anomaly Score Based on SVDD

SVDD [52], [53] was proposed to provide an accurate description of one class of objects, i.e., the target class. This method is inspired by the support vector classifier and defines a spherically shaped boundary with minimal volume around the target objects. For a target dataset $x \in R^{n\times d}$, n is the length of the dataset and d is the feature dimension; the dataset is mapped from the original space to the feature space through a nonlinear transform function, $\boldsymbol{\Phi} : x \rightarrow F$. Then, a hypersphere with the smallest volume is defined in the feature space. To construct such a minimum volume hypersphere, SVDD solves the optimisation problem in (1) [53]:

$$\begin{aligned} &\min_{a,R,\xi} \; R^2 + C\sum\nolimits_{i=1}^{n}\xi_i \\ &s.t. \;\; \left\|\boldsymbol{\Phi}\left(x_i - a\right)\right\|^2 \le R^2 + \xi_i, \quad \xi_i \ge 0, \;\; \forall i = 1, 2, \ldots, n, \end{aligned} \tag{1}$$

where R and a are the radius and centre of the hypersphere, respectively; ξ is the relaxation factor; and C is a penalty parameter that weighs the volume of the hypersphere and the error rate to prevent the model from being destroyed by individual extreme samples. In engineering practice, it is difficult to obtain a completely clean dataset which corresponds the normal operation of machines. The dual original problem in (1) is given by (2):

$$\begin{aligned} &\min_{\alpha_i} \; \sum\nolimits_{i=1}^{n}\sum\nolimits_{j=1}^{n}\alpha_i\alpha_j\boldsymbol{K}\left(x_i, x_j\right) - \sum\nolimits_{i=1}^{n}\alpha_i\boldsymbol{K}\left(x_i, x_i\right) \\ &s.t. \; 0 \le \alpha_i \le C, \quad \sum\nolimits_{i=1}^{n}\alpha_i = 1, \end{aligned} \tag{2}$$

where α_i is the Lagrange coefficient corresponding to x_i, and among all training samples; the samples whose Lagrange coefficient satisfies $0 \le \alpha_i \le C$ are called support vectors (SVs),which determine the boundary of dataset; $\boldsymbol{K}\left(x_i, x_j\right)$ is the kernel function, $\boldsymbol{K}\left(x_i, x_j\right) = \left\langle\boldsymbol{\Phi}\left(x_i\right), \boldsymbol{\Phi}\left(x_j\right)\right\rangle$. The Gaussian kernel function is considered as an example, as shown in (3):

$$\boldsymbol{K}\left(x_i, x_j\right) = \exp^{-\gamma\left\|x_i - x_j\right\|^2}, \quad \gamma = \frac{1}{2\sigma^2} \tag{3}$$

where $exp\left(\cdot\right)$ is the Gaussian kernel function, and γ is a hyperparameter that controls the width (σ) of the Gaussian function.

The calculation formulas for the centre and radius of the hypersphere are given by (4) and (5), as shown at the bottom of the next page, $x_v \in SVs$. For the test sample z, the distance from the centre of the hypersphere is calculated as (6), shown at the bottom of the next page, where $d \le R$ indicates that the test sample is on or inside the hypersphere and is thus a normal sample; otherwise, it is an abnormal sample. Fig. 2 shows the SVVD results under a banana-shaped dataset [52], [53] with different kernel function. It shows that the classifier based on Gaussian kernel function describes the smallest region with a complex boundary.

The SVDD model with Gaussian kernel function is selected to describe the vibration vectors in this work. The current research simply considers the sample as an outlier when it is outside the learned hypersphere, and discussion of the

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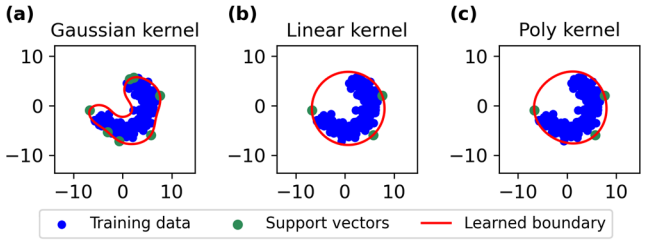


Fig. 2. SVDD results with different kernel function: (a) Gaussian kernel function; (b) Linear kernel function; (c) Polynomial kernel function.

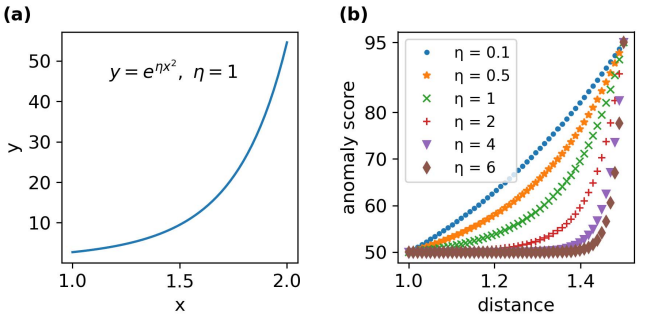


Fig. 3. The anomaly score mapping when the sample is outside the hypersphere: (a) the basis function of target mapping; (b) the target mapping with different η.

degree of deviation is lacking. This work defines the anomaly score corresponding to the degree of deviation, with a greater anomaly score indicates greater abnormal confidence. The anomaly score is set to 0 when the sample is on or inside the hypersphere. The deviation can be expressed as the distance from the border of the hypersphere. In the case of SVDD with Gaussian kernel function, the distance from the centre can be simplified as (7):

$$d = \sqrt{1 - 2\sum_{i=1}^{n} \alpha_i \boldsymbol{K}(z, x_i) + C_s}, \tag{7}$$

where $C_s = \sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j \boldsymbol{K}(x_i, x_j)$ is a constant value and is determined by the training dataset.

It is known from (3) that the Gaussian kernel function has better anti-interference ability to noise in dataset, and its parameter σ determines the action range, beyond which its action loses. That is, $\boldsymbol{K}(x_i, x_j) \geq$ Ý0 tends to be 0 when σ is fixed and $\|x_i - x_j\|^2$ is large enough. Thus, the (7) has the maximum value as shown in (8). It indicates a possible outlier, when $d \in [R, d_{max}]$.

$$d_{max} = \sqrt{1 + C_s}, \tag{8}$$

This work has two pre-sets to describes the mapping of distance and anomaly score. First, the growth rate of anomaly score increases with distance. Second, the range of anomaly score is set to 50-95, corresponding to $[R, d_{max}]$. The exponential function is selected as the basis function f_0 of target mapping in this work, as shown in Fig. 3a with $\eta = 1$.

$$f_0 : y = e^{\eta x^2}, \quad x \in [1, 2], \tag{9}$$

Define the mapping f_1, which maps the distance linearly to the definition domain of basic function f_0.

$$f_1 : \quad y = k_1 x + b_1, \quad x \in [R, d_{max}], \ y \in [1, 2], \tag{10}$$

Define the mapping f_2, which maps the range of f_0 linearly to the range of anomaly score [50, 95].

$$f_2 : \quad y = k_2 x + b_2, \quad x \in [f_0|_{x=1}, \ f_0|_{x=2}], \ y \in [50, 95], \tag{11}$$

Therefore, the target mapping $S(\cdot)$ from distance to anomaly score can be obtained. When $x > 0$, it is easy to prove that both the first and second derivatives of $S(\cdot)$ are always greater than 0.

$$S = f_2(f_0(f_1(x))) = k_2 e^{\eta(k_1 x + b_1)^2} + b_2$$
$$x \in [R, d_{max}], \quad S \in [50, 95], \tag{12}$$

From (9)-(12), the parameters in the target mapping are determined by the range of distance $[R, d_{max}]$. Assuming the distance range is [1, 1.5], the corresponding mapping is shown in Fig. 3b with different η that adjusts the change rate of the mapping. In this work, when the monitored sample falls on the surface of the hypersphere, assign its anomaly score to 50. As the degree of deviation increases, the anomaly score increases faster until it reaches to 95. The mapping shows that the growth rate of anomaly confidence increases with the distance from hypersphere. Anomaly threshold based on anomaly score has two advantages. First, the anomaly score reflects the degree of deviation, which is ignored in the original SVDD. Second, the anomaly score reflects the anomaly confidence, which can provide a robustness monitoring by different thresholds in industry application.

C. Evaluation Criteria for Single-Classifier

The SVDD is a single-classifier method which only need one target class samples in training process. Based on the model, a new sample can be evaluated whether it is belonged to target class. Thus, the result of evaluating new samples is a two-class question. The model evaluation criteria are different between training process and application process.

In the training process, only the target class data is available. The accuracy is defined as the ratio of correctly judged samples to the total samples in test dataset (from target class). K-fold cross-validation is used to avoid overfitting. The low mean accuracy indicated that the model may be overfitting,

$$a = \sum_{i=1}^{n} \alpha_i \boldsymbol{\Phi}(x_i), \tag{4}$$

$$R = \sqrt{\boldsymbol{K}(x_v, x_v) - 2\sum_{i=1}^{n} \alpha_i \boldsymbol{K}(x_v, x_i) + \sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j \boldsymbol{K}(x_i, x_j)}, \tag{5}$$

$$d = \sqrt{\boldsymbol{K}(z, z) - 2\sum_{i=1}^{n} \alpha_i \boldsymbol{K}(z, x_i) + \sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j \boldsymbol{K}(x_i, x_j)}, \tag{6}$$

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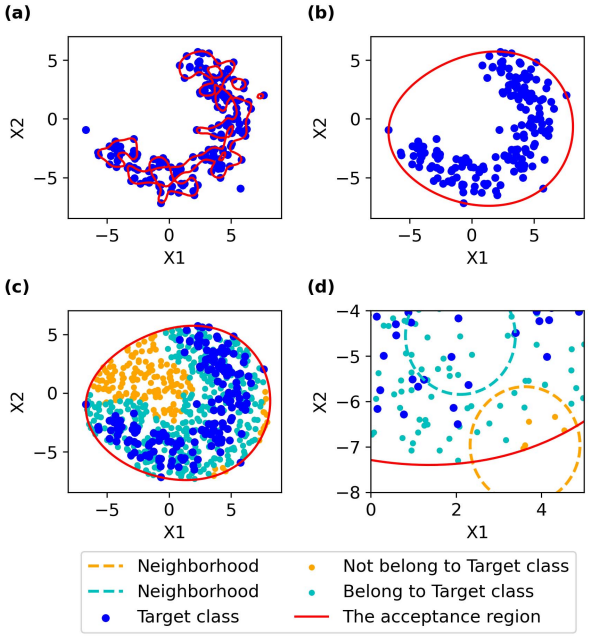


Fig. 4. The situation of overfitting and underfitting in the training process, and the principle of fit degree: (a) overfitting with mean accuracy is 0.28 and fit degree is 1.0; (b) underfitting with mean accuracy is 0.96 and fit degree is 0.68; (c) the principle of fit degree based on Monte Carlo method; (d) the approximate judgment of random samples' label.

as shown in Fig. 4a. However, in the absence of anomaly samples, too high mean accuracy may imply that the model is underfitting, that is, the model learns a large acceptance region covering all test samples, as shown in Fig. 4b.

In this work, we proposed the fit degree, which aims to characterize the coincidence degree of learned acceptance region and target class, to indicate the underfitting situation. The definition of fit degree is based on Monto Carlo method as shown in Fig. 4c. That is, the samples are randomly generated in the learned acceptance region, and the proportion belonging to the target class is defined as the fit degree. The random sample is considered to belong to the target class, if its neighborhood contains any point that belongs to the training dataset. It is significant to make a trade-off between the mean accuracy and fit degree in single-classifier model. where, T is the target class with K dimensions; S is the random samples set in the learned acceptance region and total number is M; C is the neighborhood of random sample with radius r; m is the number of random samples belong to target class.

The evaluation criteria in application process can use the confusion matrix, which aggregates records in form of a matrix according to the two criteria of the true category and predicted category, as summarised in Table I. TP is true positive, meaning that the model judges a positive class as positive; FP is false positive, meaning that the model judges a negative class as positive; FN is false negative, meaning that the model judges a positive class as negative; and TN is true negative, meaning that the model judges a negative class as negative.

The accuracy reflects the performance of the model in making correct judgments and is calculated as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

Algorithm 1 Fit Degree

Input: T, S
for $k \leftarrow 1$, K **do**
  $r_k = \max(T^k) - \min(T^k)$
end for
$r = \min\{r_k\} \ / \ 10$
$m = 0$
for $i \leftarrow 1$, M **do**
  if $\exists T_j$ in C_i, $T_j \in T$ **then**
    $m+ = 1$
  end if
end for
Return: $m \ / \ M$

TABLE I
CONFUSION MATRIX

	Predicted label	
Actual label	True Positive: TP	False Negative: FN
	False Positive: FP	True Negative: TN

The precision is for prediction results, which indicates how many of the predicted positive samples are truly positive samples. Then there are two possibilities for positive prediction. One is to predict the positive class as the positive class (TP), and the other is to predict the negative class as the positive class (FP). The precision reflects the false alarm rate in practise. When the precision is low, the false alarm rate may be high.

$$Precision = \frac{TP}{TP + FP} \tag{14}$$

The recall is for original sample, which indicates how many positive examples in the sample are predicted correctly. There are also two possibilities. One is to predict the original positive class as a positive class (TP), and the other is to predict the original positive class as a negative class (FN). The recall reflects the missing alarm rate in practise. When the recall is low, the f missing alarm rate may be high.

$$Recall = \frac{TP}{TP + FN} \tag{15}$$

In sometimes, there are contradictions between precision and recall indicators which need to be considered comprehensively. The most common method is F-score which is harmonic average of precision and recall.

$$F - score = \frac{2 * precision * recall}{precision + recall} \tag{16}$$

D. Proposed Anomaly Detection Method

The vibration monitoring is a common technology for condition monitoring that is of great significance for safe and reliable running of rotating machinery. However, the fixed threshold-based methods for monitored indicators such as amplitude, are applied in on-line monitoring of real production, in which thresholds are man-made according to experience.

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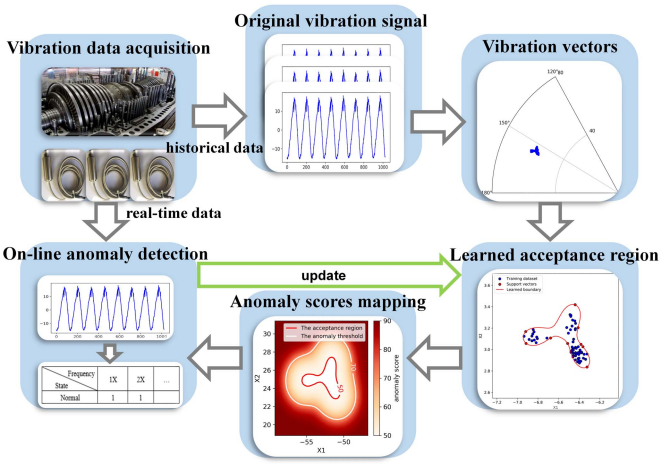


Fig. 5. Flowchart of the proposed anomaly detection method for rotating machinery.

The fixed threshold has an obvious shortcoming which causes either frequent alarms with a low threshold or missing of best maintenance opportunity with a high threshold. The traditional vibration monitoring often ignores the phase changes at different CF. When the rotating machinery is in the normal operation state, we have found that the 1X vibration vector is distributed in a certain region in the polar plot, and the vibration vector with fault deviates from this region. Based on this motivation, an intelligent and automatic anomaly detection method that considers the amplitude and phase information jointly is proposed for fault early warning. The method is divided into six steps, as shown in Fig. 5.

Step 1: Vibration data acquisition. The proposed method utilises the vibration data from the condition monitoring process for rotating machinery. The obtained dataset is divided into offline and online dataset. Offline dataset is used to learn the acceptance region of the vibration vector for each CF, and online dataset is used to evaluate the equipment status in real-time. The data acquisition meets the FFT processing requires.

Step 2: Original vibration signal. The recorded vibration signal data is pre-processed, including noise reduction and smoothing, to obtain the original vibration waveform. Each waveform is a sample that contains the current vibration information of the target equipment.

Step 3: Vibration vectors. FFT-OA technology is used to extract information from the vibration waveform, and a series of vibration vectors corresponding to CFs (0.5X, 1X, 2X…) are obtained. The vibration vectors contain the amplitude and phase information and gather in a certain region when the equipment in normal operation.

Step 4: Learned acceptance region. The amplitude and phase have different numerical characteristic which amplitude range is greater than zero whereas the phase varies from 0 to 2π. The conversion is given in (17), where $[X1, X2]^T$ is the new vibration vector in Cartesian generated from $[A, \varphi]^T$ in the polar plot. The Euclidean distance between any two points does not change with the conversion from polar to Cartesian plot. Moreover, the conversion has a better visualization for the distribution of vibration vectors and simplifies the calculation of the constraint function in the SVDD model.

$$[X1, X2]^T = \begin{bmatrix} cos\varphi & 0 \\ 0 & sin\varphi \end{bmatrix} [A, A]^T \tag{17}$$

where, $[A, \varphi]^T$ is a vibration vector in Polar plot, A is the amplitude, φ is the phase. $[X1, X2]^T$ is the new vibration vector in Cartesian, X1 and X2 have similar numerical characteristic.

Step 5: Anomaly score mapping. For the trained SVDD model, the anomaly score mapping program is determined by the mapping (12). For a certain CF, the anomaly score is 0 with the vibration vector is inside the acceptance region, the anomaly score is mapped from 50-95 with the vibration vector is outside the region. It is judged as the possible abnormal and provides an alarm with anomaly score. In the monitoring program, a threshold based on anomaly score can be set to report alarm, e.g.,70 was set in this work.

Step 6: On-line anomaly detection. Based on the trained SVDD model, the real-time vibration data is processed in Steps 2–3 to assessment whether the vibration vector is within the acceptance region or not for each CF. In actual production, the anomaly threshold can be set by anomaly score which corresponds to anomaly confidence, the higher score the higher confidence for anomaly alarm. The alarm result is confirmed by the operation and maintenance personnel. If the alarm is true, the maintenance activities shall be arranged. If the alarm is wrong, the SVDD model shall be updated.

In this method, the amplitude and phase of each CF are considered jointly for the vibration signal. From the extensive historical vibration signal data, the acceptance region of vibration vector can be learned for each CF, which provides the evaluation basis for anomaly detection. Compared with the early warning method based on fixed amplitude threshold, the proposed method has three advantages: First, it can make full use of the information from the vibration signal data, including amplitude and phase. Second, the anomaly determination criteria come from the description of normal data rather than human experience. Third, this method can be extended to any N-order CF, so that can provide a more comprehensive vibration-based condition monitoring for the machinery.

III. Case Studies

A. Constant Speed Operation Condition

In thermal power plant, the steam turbine connected with generator usually works in constant speed operation (CSO) condition. The vibration dataset from a real steam turbine is used to validate the proposed method. The unit is an LCH-1100-27 Ultra Supercritical jointly produced by the Harbin Turbine Power Generation Group Co., Ltd. and Toshiba Company in Japan, consisting of one intermediate reheat, four cylinders and four rows of steam, a single axle impulse type, and double back pressure condensing type. A schematic diagram of the shafting support is shown in Fig. 6, with bearings #1–#10.

According to unit operation and maintenance records, a rubbing fault case in #7 bearing is selected. In this study, 178 sets of normal operation vibration signals and 74 sets of early abnormal vibration signals were obtained when the speed

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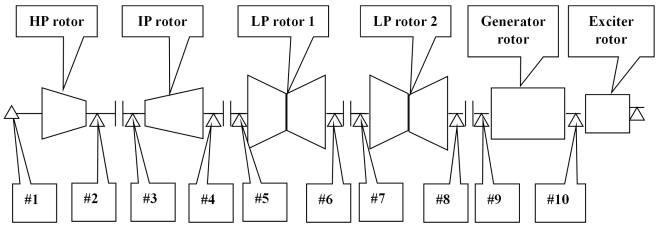


Fig. 6. Schematic diagram of the shafting support in steam turbine.

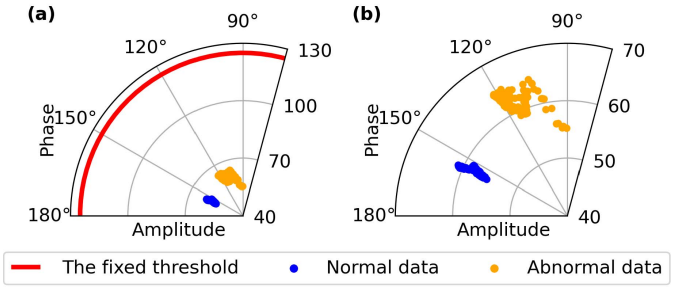


Fig. 7. The obtained vibration dataset on a constant speed operation case: (a) 1X vibration vectors in Polar plot with normal and abnormal data; (b) the detailed differences between normal and abnormal data.

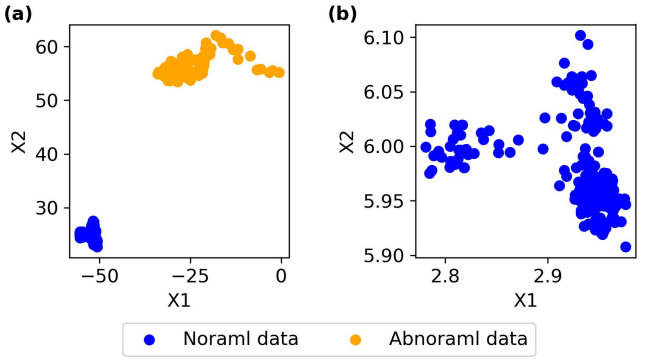


Fig. 8. The vibration vectors in Fig. 7 are mapped from Polar to Cartesian plot: (a) 1X vibration vectors in Cartesian plot with normal and abnormal data; (b) the normal vibration vectors after normalization.

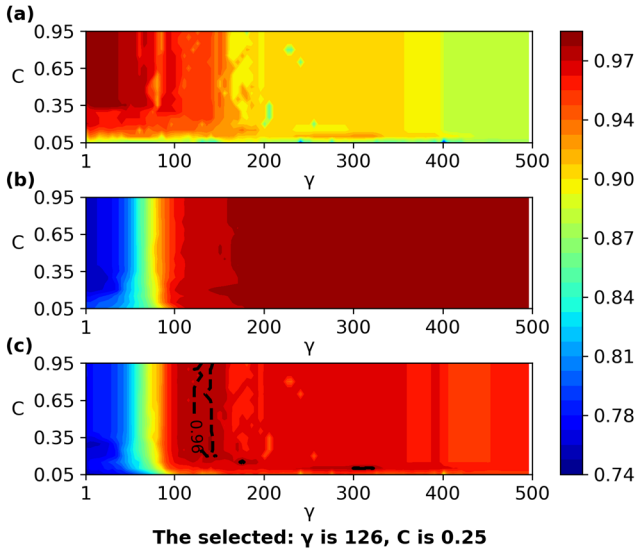


Fig. 9. Grid search for γ and C in SVDD model on a constant speed operation case: (a) grid search on mean accuracy; (b) grid search on fit degree; (c) the weighted average of mean accuracy and fit degree.

of turbine was 3000 rpm. The vibration signal sampling frequency is 128 points/revolution, and each set of vibration signal contains 1024 points. Fig. 7 shows the obtained vibration dataset on CSO condition, which contains normal data and abnormal data with early stage rubbing fault. In Fig. 7a, the red line is fixed-threshold based on amplitude, which cannot detect the early fault. By contrast, the difference between the normal and early anomaly states is obvious when phase is considered in Polar plot. The normal samples are clustered in a certain region, and the abnormal samples are dispersed and deviate from normal region. Thus, it is feasible to detect anomalies by learning the acceptance region of normal samples.

Due to the different numerical characteristic between amplitude and phase. The vibration vectors were mapped from Polar to Cartesian plot for better visualization and simplified calculation. Fig. 8a illustrates the mapping, it can be seen that the distinction between normal and abnormal is clear as seem as in Fig. 7. In this case, the fixed amplitude threshold in production is 125 μm, we set the range of X1 and X2 is $[-125, 125]^T$. The X were normalized to $[0, 10]$ for optimization of the hyperparameters of the SVDD model under the unified scale.

The total 178 sets of normal samples were divided into two parts, 100 sets were used to learn the normal boundary. The rest 78 sets of normal samples and total 74 sets of abnormal samples composed the test dataset. A grid search was performed on hyperparameters γ and C of SVDD model training with mean accuracy and fit degree as the optimisation objectives, and the result is shown in Fig. 9. With the increase of γ, the mean accuracy changes from large to small and tends to be stable, the fit degree changes from small to large and tends to be stable. The fit degree can effective indicate the underfitting model with small γ. The C has little influence on the model, and possible reason is that the training dataset is relatively concentrated with small noise. In this work, the weighted average of mean accuracy and fit degree was taken as the performance indicator as shown in Fig. 9c. Among the top 10 values of the performance indicator, the best combination of hyperparameters is the one with the highest mean accuracy. The selected γ is 126 and C is 0.25 in this case study.

The SVDD, minimum covariance determinant (MCD) [54], and isolated forest (iForest) [55] methods are used to learn the normal boundary as shown in Fig. 10a, and the results based on acceptance region are compared in Table II. The result shows that the data description models are trained with low time cost. The MCD is based on the covariance between features and tends to learn an elliptical region and the error is large when the data distribution is irregular. The iForest model becomes unstable when there are few training samples or random trees and tends to work on the target class with high density. The SVDD describes the acceptance region more accurately and reasonably than the other two methods, which demonstrates that SVDD can better describe the distribution of vibration vectors under normal conditions. Fig. 10b shows

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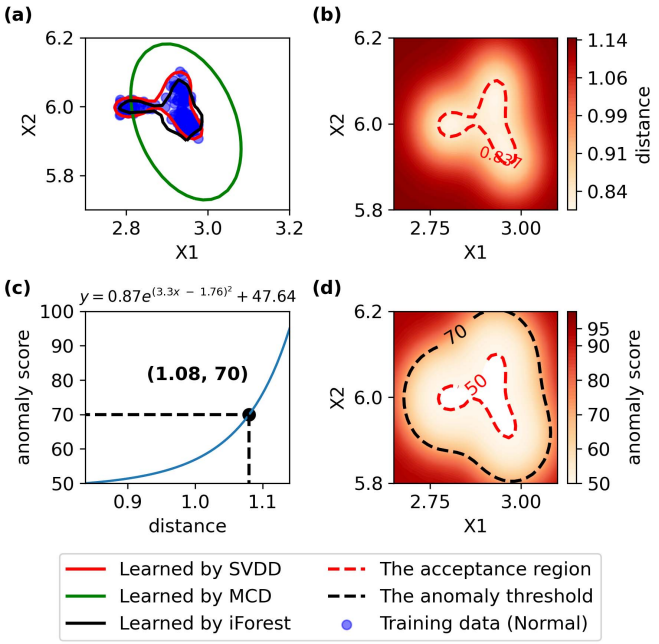


Fig. 10. The learned acceptance region and anomaly score mapping on a constant speed operation case: (a) the learned acceptance region by three methods; (b) the detail acceptance region learned by SVDD; (c) the anomaly score mapping based on SVDD model; (d) the tolerance of alarms based on anomaly threshold.

TABLE II
RESULTS OF DIFFERENT DATA DESCRIPTION METHODS UNDER THE A CONSTANT SPEED OPERATION CASE

Confusion matrix			Evaluation indicators
SVDD	Predicted label		Time cost: 0.054 s Precision: 1.0 Recall: 1.0 Accuracy: 1.0 F-score: 1.0
Actual label	74	0	
	0	78	
MCD	Predicted label		Time cost: 0.030 s Precision: 0.902 Recall: 1.0 Accuracy: 0.947 F-score: 0.949
Actual label	74	0	
	8	70	
iForest	Predicted label		Time cost: 0.093s Precision: 0.948 Recall: 1.0 Accuracy: 0.974 F-score: 0.973
Actual label	74	0	
	4	74	
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the SVDD model can provide hierarchical alarm by different threshold of distance from center. According to (12) and set $\eta = 1$, the anomaly score was mapped from the distance to provide an intuitive anomaly confidence, as shown in Fig. 10c. There is a tolerance between outside the acceptance region and alarm, which makes the model more robust in application. The higher anomaly threshold, the higher tolerance of the data description model. As shown in Fig. 10d, the anomaly threshold was set to 70 in this work.

Compared with amplitude-based method in Fig. 7, the vibration vector-based method can detect the early anomalies as shown in Fig. 11 and is meaningful to safe and reliable

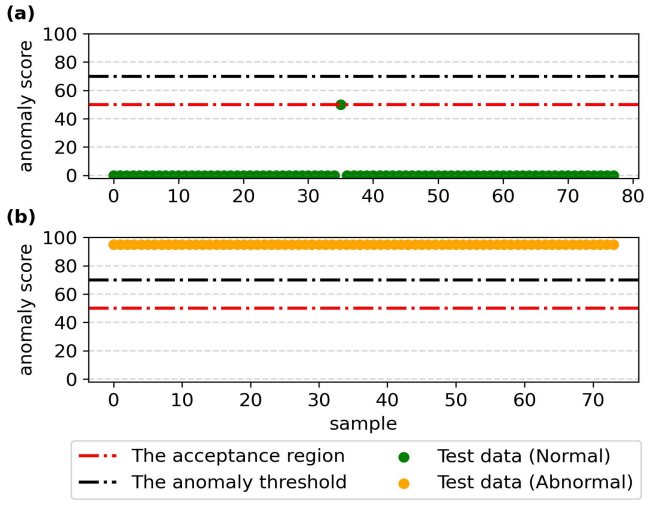


Fig. 11. The test result of normal and abnormal data on a constant speed operation case by SVDD model: (a) the anomaly score of normal test data; (b) the anomaly score of abnormal test data.

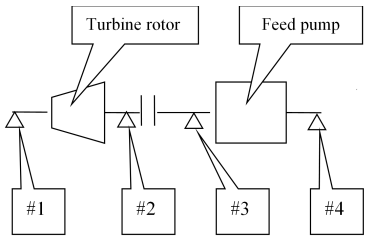


Fig. 12. Schematic diagram of the shafting support in steam feed pump.

production in industry. The anomaly score indicates the degree of deviation from the acceptance region. In this case study, anomaly scores of abnormal samples are 95, which means that these samples are beyond the action scope of the support vector used to describe the acceptance region. Hence, the confidence of the anomaly is high.

B. Variable Speed Operation Condition

The steam turbine is a large-scale rotating machine that typically operates at a constant speed. In practice, rotating machinery is also frequently used in variable speed operation (VSO) conditions. To verify the generality of the proposed method, a variable speed dataset was obtained from a steam feed pump. A schematic diagram of the shafting support is shown in Fig. 12, with bearings #1–#4. The steam feed pump is an important auxiliary equipment in thermal power plant, which operates at variable speed.

According to unit operation and maintenance records, a rubbing fault case in #2 bearing is selected. In this case study, 1593 sets of normal operation vibration signals and 374 sets of early abnormal vibration signals were obtained. Due to the limited data sources, the obtained dataset covers samples with speed of 4600-5200rpm. The vibration signal sampling frequency is 128 points/revolution, and each set of vibration signal contains 1024 points. Fig. 13 shows the obtained vibration dataset on VSO condition, which contains normal data and

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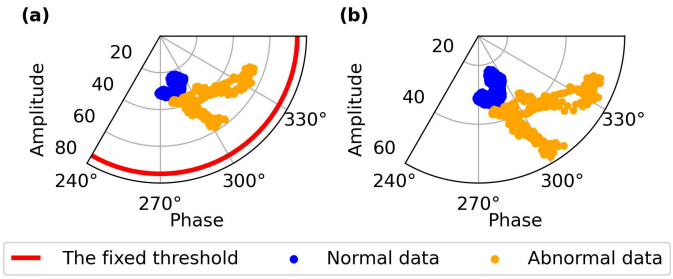


Fig. 13. The obtained vibration dataset on a variable speed operation case: (a) 1X vibration vectors in Polar plot with normal and abnormal data; (b) the detailed differences between normal and abnormal data.

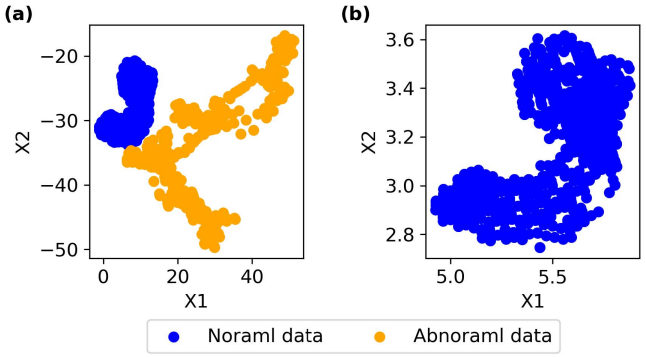


Fig. 14. The vibration vectors in Fig. 13. are mapped from Polar to Cartesian plot: (a) 1X vibration vectors in Cartesian plot with normal and abnormal data; (b) the normal vibration vectors after normalization.

abnormal data with early stage rubbing fault. The difference between the normal and early anomaly states is obvious when phase is considered in Polar plot as shown in Fig. 13 a. The red line is fixed-threshold based on amplitude, which cannot detect the early fault. The normal samples are clustered in a certain region, and the abnormal samples are dispersed and deviate from normal region. Thus, it is practicable to detect anomalies by learning the acceptance region of normal samples.

The vibration vectors were mapped from Polar to Cartesian plot as shown in Fig. 14a, and it can be seen that the distinction between normal and abnormal is clear as seem as in Fig. 13, and the normal region can be learned to detect anomalies. The fixed amplitude threshold in production is 75 μm in this case study. We set the range of X1 and X2 is $[-75, 75]^T$, the X were normalized to [0,10] for optimization of the hyperparameters of the SVDD model under the unified scale.

The total 1593 sets of normal samples were divided into two parts, 1216 sets were used to learn the normal boundary. The rest 377 sets of normal samples and total 374 sets of abnormal samples composed the test dataset. The grid search result of mean accuracy and fit degree in SVDD model training process is shown in Fig. 15. As the same, with the increase of γ, the mean accuracy changes from large to small and tends to be stable, the fit degree changes from small to large and tends to be stable. The fit degree also can indicate the underfitting model with small γ. The C has little influence on the model, and the possible reason is that the training dataset is relatively

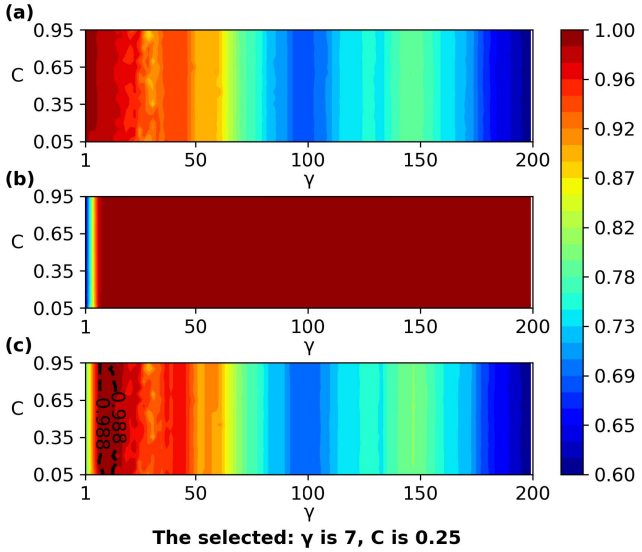


Fig. 15. Grid search for γ and C in SVDD model on a variable speed operation case: (a) grid search on mean accuracy; (b) grid search on fit degree; (c) the weighted average of mean accuracy and fit degree.

TABLE III
RESULTS OF DIFFERENT DATA DESCRIPTION METHODS UNDER THE A VARIABLE SPEED OPERATION CASE

Confusion matrix			Evaluation indicators
SVDD	Predicted label		Time cost: 2.84 s Precision: 1.0 Recall: 1.0 Accuracy: 1.0 F-score: 1.0
Actual label	374	0	
	0	377	
MCD	Predicted label		Time cost: 0.848 s Precision: 0.933 Recall: 1.0 Accuracy: 0.964 F-score: 0.965
Actual label	374	0	
	27	350	
iForest	Predicted label		Time cost: 0.224 s Precision: 0.919 Recall: 1.0 Accuracy: 0.956 F-score: 0.958
Actual label	374	0	
	33	344	
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concentrated with small noise. The weighted average of mean accuracy and fit degree tends to be high when γ is less than 50. Among the top 10 values of the weighted average, the best combination of hyperparameters is the one with the highest mean accuracy. The selected γ is 7 and C is 0.25 in this case study. Due to the magnitude difference in the range of data sets, the selected γ is also different.

Similarly, three methods (SVDD, MCD, and iForest) were used to obtain the acceptance region of vibration vector with the normal samples, and the results are compared in Table III. As shown in Fig. 16a, the MCD tends to learn an elliptical region and with a low fit degree, the iForest tends to learn a smaller region than SVDD, because the training dataset in this case study has a low density at the boundary. In Fig. 16b, the SVDD model can provide hierarchical alarm by different threshold of distance from center. According to (12) and set $\eta = 1$, the anomaly score was mapped from the distance to

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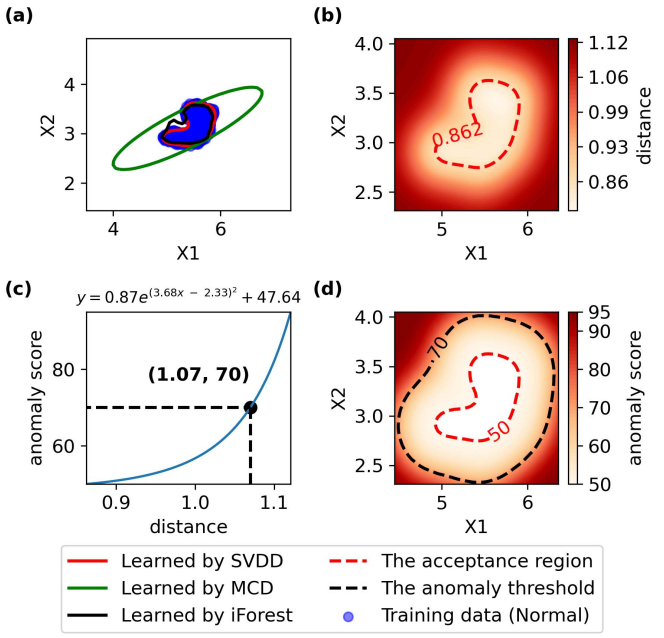


Fig. 16. The learned acceptance region by three methods on a variable speed operation case: (a) the comparison of acceptance region by SVDD,MCD, iForest; (b) the detail acceptance region learned by SVDD; (c) the anomaly score mapping based on SVDD model; (d) the tolerance of alarms based on anomaly threshold.

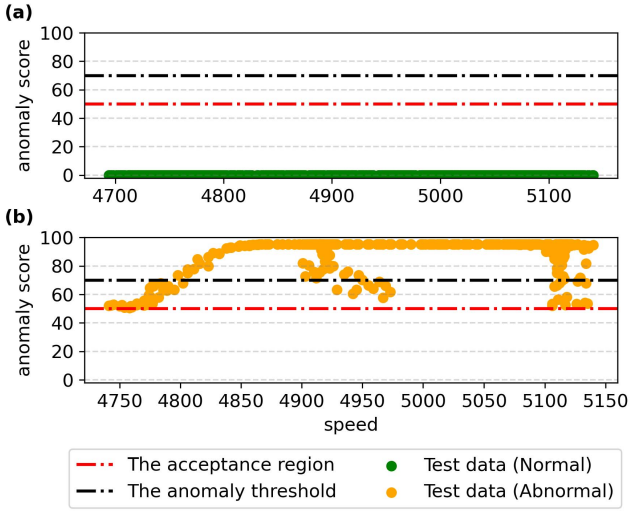


Fig. 17. The test result of normal and abnormal data on a variable speed operation case by SVDD model: (a) the anomaly score of normal test data; (b) the anomaly score of abnormal test data.

provide an intuitive anomaly confidence, as shown in Fig. 16c. To make the model more robust in application, there is a tolerance between outside the acceptance region and alarm. The higher anomaly threshold, the higher tolerance of the data description model. The anomaly threshold was set to 70 in this work, as shown in Fig. 16d. The anomaly score indicates the degree of deviation from the acceptance region.

Compared with amplitude-based method in Fig. 13, the vibration vector-based method can detect the early anomalies with a high recall as shown in Fig. 17, which is important for health monitoring in industry. In this case study, the speed range is 4600-5200rpm, the confidence of anomaly becomes high with the increase of rotating speed. The abnormal samples exceed the action range of the support vector describing the acceptance region and the anomaly score are 95, when the rotating speed reaches 4850rpm.

IV. Conclusion and Future Work

This work proposes an intelligent anomaly detection method for rotating machinery based on vibration vector in target characteristic frequency, which is verified by two practice case and has a prospect to the edge computing for machine condition monitoring. The vibration vector joints the amplitude and phase information, can provide more effective anomaly detection than amplitude-based method in sometimes. For the single-classifier training process, the K-fold cross-validation was applied to avoid overfitting, and the fit degree was proposed and defined to indicate underfitting. There is a trade-off between mean accuracy and fit degree, to obtain a robust data description model in industry application. The minimum covariance determinant tends to learn an elliptical region, and the fit degree is low when the target class does not obey the Gaussian distribution. The learned acceptance by isolated forests is unstable, which is related to the data density of target class. The support vector data description method does not need a priori knowledge of the target class to learn the acceptance region, which can update easily and rapidly in real time. Base on the support vector data description with Gaussian kernel function, the anomaly score was defined to indicate the degree of deviation from the acceptance region, which provides a robustness monitoring in industrial applications.

With the improvement of O&M requirements, anomaly detection based single-classifier may be more utilitarian for two reasons in the fault early warning, which aims to reduce the cost. First, it is easy to obtain a large amount of data under normal operating conditions, which avoids the possible tedious data collection in multi-classifier. Second, the natural deterioration over the entire lifecycle may make the monitored features change, and the single-classifier can be updated easily with current normal dataset. Besides the vibration vector, many vibration feature extraction methods have been proposed from various aspects, the concise and useful feature is one of the goals. There may be better solution to express the underfitting of single-classifier model. In addition, more early stage fault cases are needed to validate the proposed method at the slow rotational speeds in real production.

Acknowledgment

The authors would like to thank Xiaofeng He at Jiangsu Fangtian Electric Power Technology Company Ltd., for his support and contribution in providing vibration signal datasets form a practice turbine.

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